Problem FoL.1 ... curtain

Imagine a thin rigid homogeneous rod with mass $m = 1\,\mathrm{kg}$ *and length* $l = 2\,\mathrm{m}$*, which is attache*
to a horizontal rail by a small massless ring at its end, so that it can slide without friction. A
the ring accelerates at a constant rate $a = 5\,\mathrm{m{\cdot}s^{-2}}$*, the rod makes a constant angle* φ *to th*
vertical. Find this angle given that the whole situation takes place on the Earth's surface, thu
in the presence of the vertical acceleration due to gravity $g = 10\,\mathrm{m{\cdot}s^{-2}}$*, neglecting the effects o*
air resistance. State your answer in radians. *Náry, looking at the mechanics of curtain*

Problem FoL.2 ... hangmen

There are two balls of mass m and electric charge q (with the same sign) hanging from two sling
of length l fixed in the same point. These balls are placed in air with density $\varrho_{\mathrm{a}} = 1.2\,\mathrm{kg{\cdot}m^{-3}}$
Due to the electric repulsion of the balls, the slings are forming an angle α. If we put the sam
parate in the olive oil with relative permitivity $\varepsilon_{\mathrm{r}} = 3$ and density $\varrho_o = 900\,\mathrm{kg{\cdot}m^3}$, the angl
will remain the same. Consider the permitivity of the air to be same as permitivity od vacuum
What is the density of the balls?
f(Aleš) was reading about little balls and decided to recompute the problem for nicer situation

Problem FoL.3 ... blue rock

Center of mass of a mountaineer climbing on a rock is $h = 24\,\mathrm{m}$ *above ground. The last belay* *place where the climber's rope runs through a metallic circle attached to the rock) is at a height* $_0 = 20\,\mathrm{m}$. *The climber slips and falls. How closest to the ground does he get during his fall?* *Young's modulus of the rope is* $E = 100\,\mathrm{MPa}$, *its radius* $r = 0.5\,\mathrm{cm}$ *and mass of the climber* $n = 70\,\mathrm{kg}$. *Neglect mass of the rope and all friction. Assume that the rope is attached to the* *limber in his center of mass. All distances are given with respect to a securing device which is* *ttached to the ground and does not move during the fall. The local gravitational acceleration* *s* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Honza thought carefully before climbing

Problem FoL.4 ... gas block

There is a movable divider in a closed cylindrical vessel, separating it into two chambers. On of the chambers contains 25 mg *of* N_2 *while the second one contains* 40 mg *of* He. *Assume tha he equilibrium state was attained. What is the ratio of the lenghts of the chambers in th equilibrium state? Assume ideal-gas behaviour. Your answer should be less than 1.*

Janapka was going through her old exercise books

Problem FoL.5 ... marble play

We want to galvanise (in copper sulphate solution) an iron sphere of mass $m_{\mathrm{Fe}} = 8\,\mathrm{kg}$ *and density* $\varrho_{\mathrm{Fe}} = 8\,\mathrm{g{\cdot}cm^{-3}}$. *We aim for a layer of copper* $\Delta r = 2\,\mathrm{mm}$ *thick. The density of copper is* $\varrho_{\mathrm{Cu}} = 9\,\mathrm{g{\cdot}cm^{-3}}$, *molar mass is* $M_{\mathrm{Cu}} = 63.5\,\mathrm{kg{\cdot}mol^{-1}}$ *and we apply constant current* $I = 0.5\,\mathrm{A}$. *How long does the process take? State your answer in days, ceiled (i.e.* 3.238 *days are* 4 *days).*

Kiki was remembering the chemistry olympiad

Problem FoL.6 ... springy one

We happen to have forgotten a spring with natural length $l = 0.5\,\mathrm{m}$ and a spring constan
of $k = 5\,\mathrm{N{\cdot}m^{-1}}$ in the outer space. There is a mass $m_1 = 1\,\mathrm{kg}$ attached to one end of the sprin
while at the other end, there is a mass $m_2 = 3\,\mathrm{kg}$. Compute the period of small oscillations o
the spring.

Paťo was going through his old exercise book

Problem FoL.7 ... expandable balloon

A closed helium balloon is taking off from the Earth's surface, where the temperature an pressure are 300 K *and* 101 kPa *respectively. Eventually, it will reach the point where th temperature and pressure are* 258 K *and* 78 kPa *respectively. Assuming that the balloon is o spherical shape initially with radius* 10 m*, find the factor by which its radius will increase. Th balloon is in thermodynamic equilibrium with its environment. Do not take into account th surface tension of the balloon.* *Tomáš was dreaming about flying in a hot air balloon*

Problem FoL.8 ... resistant prism

The edges of a square pyramid are made out of wires which are conductively connected at a vertices. Compute the resistance across the opposite vertices on a diagonal of the base square given that the resistance of one metre of the wire is $1\,\Omega$*, the height of the pyramid is* $\sqrt{7}\,\mathrm{m}$ *an the base length is* $2\,\mathrm{m}$*.*

Pikoš, while marking solutions

Problem FoL.9 ... satellite

Consider a planet and its natural satellite orbiting about their common centre of mass, wher the motion takes place in a plane. The magnitude of the tangent velocity of the satellite, define with respect to the centre of mass of the system, is $2.5\,\mathrm{km{\cdot}s^{-1}}$*. Find the ratio of the mass of th planet to the mass of the satellite in order for the centre of mass of the system to be locate at the planet's surface, given that the orbits are circular. The mass of the planet is* M_p = $= 7.6 \cdot 10^{24}\,\mathrm{kg}$*, it has a radius of* $R_\mathrm{p} = 7\,436\,\mathrm{km}$ *and the radius of the satellite is* $R_\mathrm{m} = 1\,943\,\mathrm{km}$

Nicola was thinking about the two-body problem

Problem FoL.10 ... a loop

Consider an electron-emitting source with an emitting speed of $v = 1.5 \cdot 10^7\,\mathrm{m \cdot s^{-1}}$. *At a point* P *the electrons enter a homogeneous magnetic field with a magnitude of* $B = 1 \cdot 10^{-3}\,\mathrm{T}$. *The vector of electrons' velocity at* P *makes an angle* $\varphi = 15^\circ$ *to the magnetic field vector. Find the distance of* P *from the point where the electrons again (for the first time) cross the field line going through* P.

Zdeněk has teleported inside a monitor

Problem FoL.11 ... ecologically passive

A passionate tree hugger (weighing only $m = 50\,\mathrm{kg}$) learned that the city council decided t
chop down his favourite tree. He climbed onto the top of his homogeneous green friend believin
that he would keep the tree killers away. However, the lumberjacks came and cut down the h =
= $10\,\mathrm{m}$ tall tree weighing $M = 1\,\mathrm{t}$. What was the speed with which the tree hugger, initiall
esting at the tree top, hit the ground? The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$

Terka, while climbing a tree

Problem FoL.12 ... nucleus on a diet

Imagine we have a nucleus with nucleon number $A = 36$, proton number $Z = 17$ and the mas
of $m_k = 5.99965 \cdot 10^{-26}$ kg What would be the sum of bonding energies in nucleus in one mol
of nuclei of this element? Provide your answer in TJ.

Kiki was bored during the lecture on anorganic chemistry

Problem FoL.13 ... music of the locomotives

Two locomotives A and B are moving with velocities $v_A = 15\,\mathrm{m{\cdot}s^{-1}}$ in the right directio and $v_B = 30\,\mathrm{m{\cdot}s^{-1}}$ in the left direction, facing each other on the paralel railways. Locomotiv A whistles on a frequence $200\,\mathrm{Hz}$. The speed of the sound is $340\,\mathrm{m{\cdot}s^{-1}}$. Let's assume tha some of the sonic waves will be reflected from the locomotive B back to locomotive A. Whic frequency will be heard by the engineer in the locomotive A? Assume that rightwards directio is positive and the environment is not moving.

Janapka was playing with trains

Problem FoL.14 ... capacitors in the circuit

Voltage in the circuit on the attached image is 10 V *and each capacitor has capacity* 10 μF
Compute the charge (in μC*) on the capacitor* C_1*.* *Dominika on a branch*

Problem FoL.15 ... glasses or contact lenses

Pepa needs glasses with focal length $f_\mathrm{b} = 50\,\mathrm{cm}$ to clearly see his favourite crossword a distance $D = 25\,\mathrm{cm}$ from his eye lens. The glasses are at $d = 2\,\mathrm{cm}$ from his eye lens whic itself has focal length $f_\mathrm{o} = 2\,\mathrm{cm}$. What focal distance (in cetimetres) must Pepa's contact lense – which are in direct contact with the eye – have so that Pepa still could see the crosswor clearly without any change in the focal length of the eye lens? Treat all lenses as thin.

Lukáš stared into crosswords

Problem FoL.16 ... superluminal electron

Assume the Bohr model of an ionized atom with fixed nucleus (with proton number Z) and onl one electron. Which would be the lowest possible proton number of the atom in order to obtai superluminal velocity of the orbiting electron? Assume that the electron is in a ground state Speed of light is $c = 299.8 \cdot 10^6\,\mathrm{m{\cdot}s^{-1}}$, charge of the electron is $e = -1.6022 \cdot 10^{-19}\,\mathrm{C}$, Coulomb' constant is $k_\mathrm{e} = 8.987 \cdot 10^9\,\mathrm{N{\cdot}m^2{\cdot}C^{-2}}$ and the reduced Planck constant is $\hbar = 1.0546 \cdot 10^{-34}\,\mathrm{J{\cdot}s}$ Use classical, not relativistic, physics.

Jakub wanted to destroy the world

Problem FoL.17 ... ecocar

Find the mass of a compressed gas which is equivalent to one litre of petrol with a heatin
value of $L = 30\,\mathrm{MJ{\cdot}l^{-1}}$, *given that we use air as our working gas with an initial pressure o*
$p_0 = 200 p_\mathrm{a}$, *where* $p_\mathrm{atm} = 10^5\,\mathrm{hPa}$ *is the atmospheric pressure. Further assume that the densit*
of air under atmospheric pressure is $\varrho_\mathrm{atm} = 1.3\,\mathrm{kg{\cdot}m^{-3}}$, *that it exhibits ideal-gas behaviour an*
that the expansion process is carried out isothermally without any loss of energy.

Lukáš came up with this on the trip in the mountains

Problem FoL.18 ... swinging barrel

A cylindrical object with a radius of $r = 0.5\,\mathrm{m}$, *height* $l = 3\,\mathrm{m}$ *and mass of two tonnes is floatin*
on the water of density ϱ *so that the axis of the cylinder remains vertical. Let us displace th*
cylinder from its equilibrium position vertically by $\Delta x = 1\,\mathrm{mm}$. *Find the period of oscillation*
of the cylinder (in seconds). The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Pikoš's problem from the school days

Problem FoL.19 ... capacitor duo

Find the charge in Coulombs on the capacitor C_2 if you know the following: At the beginning the switch S_0 was off and the switch S_1 was on, as displayed on the picture. There was zero voltage on both capacitors. Then we switched S_0 on and waited until the current stopped flowing. Subsequently, we switched S_1 off and again waited for the circuit to come into a steady state. At the end, we measured the charge on the capacitor C_2. The voltage across the ideal voltage source (DC) is $U = 17\,\mathrm{V}$ and both capacitors have the same capacitance $C = 1\,\mu\mathrm{F}$.

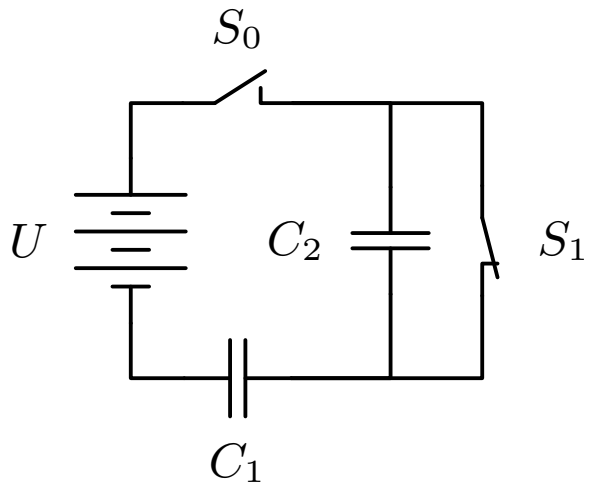


Fig. 2: To the problem 19

Náry felt like doing some electrotechnics

Problem FoL.20 ... heureka

Consider a cube with edge length $a = 1\,\mathrm{m}$ and density $\varrho_0 = 1\,000\,\mathrm{kg{\cdot}m^{-3}}$ and a container wit a liquid of density (at its level) ϱ_0. The density of the liquid increases linearly with depth to $\varrho(h) = \varrho_0 + \alpha h$, where $\alpha = 25\,\mathrm{kg{\cdot}m^{-4}}$. How deep does the cube sink, given that the height o the level of the liquid does not change having immersed the cube in the liquid? The acceleratio due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$. State your answer in centimetres.

Terka was bathing in strange liquids

Problem FoL.21 ... ecology above all

Find the efficiency (energy returned on energy invested - ERoEI) of storing the energy in th
orm of compressed air. If there is surplus of electrical energy produced in solar power plants
ve start to compress the air adiabatically so that we eventually achieve a compression ratio c
$= 10$. *If we need the energy back, we let the air expand adiabatically to the original pressure*
However, before we commenced the expansion process the air cooled down to the temperatur
efore the compression took place. Assume that the air is a diatomic gas exhibiting ideal-ga
roperties. State your answer in terms of percentage.

Lukáš was listening to a programme about the cars running on compressed air

Problem FoL.22 ... resistor blackbox

Consider a box containing three resistors with unknown resistances. These resistors are con
nected through ideal conductors. There are four output terminals leading outside the box. W
measure the resistance across different pairs of terminals for all possible combinations of ter
minals. Five such measurements yield $9\,\Omega$, $12\,\Omega$, $8\,\Omega$, $3\,\Omega$ *and* $14\,\Omega$. *Let us now connect th*
ohmmeter across the last unmeasured pair of terminals. What resistance do we find?

Náry is tragic

Problem FoL.23 ... orbiters

Determine the magnitude of a magnetic field which is present at the centre of mass of th system of two planets charged with equal charges of $Q = 100\,\mathrm{TC}$ *and with equal masses o* $M = 5 \cdot 10^{24}\,\mathrm{kg}$ *orbiting each other at a distance of* $d = 500\,000\,\mathrm{km}$ *apart. In your calculations egard the two planets as point charges and assume that their orbits are circular. State you nswer in terms of* nT. *Lukáš wanted to create an unconventional problem no matter wha*

Problem FoL.24 ... mercujet

Consider a flask filled with a potion of density $\varrho = 13\,600\,\mathrm{kg{\cdot}m^{-3}}$. The flask is sealed a the top, with a small capillary inserted in the seal. The potion is heated to the temperatur corresponding to its boiling point under the atmospheric pressure $p_\mathrm{a} = 101\,325\,\mathrm{Pa}$. By hov much will the level of the potion in the capillary rise? Neglect both the change in the surfac ension with the temperature as well as the thermal expansion of both the potion and the flask The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$. The flask is placed in vacuum. Assume tha t the beginning of the process, the partial pressure of the potion vapor is zero.

Lukáš was playing with a PET bottle

Problem FoL.25 ... dimensionless hydrogen

Consider the ground state energy of the hydrogen atom (a system consisting of a proton and an electron) in the non-relativistic model with the proton fixed at a point. If the electron i

n the infinite distance from the proton, the energy of the configuration is defined to be zero
Find the ground state energy in terms of dimensionless units defined by putting the mass o
he electron $m_e = 1$, reduced Planck constant $\hbar = 1$ and $k_e e^2 = 1$, where k_e is the Coulomb'
onstant and e denotes the electron's charge. *Jakub was forced to thin*

Problem FoL.26 ... jumping dog

Imagine a marble placed in a height of $h = 1\,\mathrm{m}$ above the ground. At some point, we releas the marble and simultaneously start to push the ground towards the marble with a speed o $V = 1.2\,\mathrm{m{\cdot}s^{-1}}$ with respect to the initial rest frame of the system. Given that the coefficient o restitution between the marble and the ground is $e = 0.3$, find the time needed for the marbl to steady on the moving ground. We define the coefficient of restitution as the ratio of speed of the marble with respect to the ground before and after each collision.

Kuba played with a marble

Problem FoL.27 ... P5

What is the average number of photons arriving from Pluto's spherical moon P5 onto the mirro of the Hubble Space Telescope every second? Assume the following: the moon has a diamete of $D = 20\,\mathrm{km}$, its distance from the Sun is $L = 32\,\mathrm{AU}$ and it has an albedo of $a = 0.3$. Th Hubble Space Telescope is equipped with a mirror of diameter $d = 2\,\mathrm{m}$. You can regard th Sun as being a monochromatic light source, emitting at a wavelength of $\lambda = 550\,\mathrm{nm}$ with th solar constant being $P_\mathrm{S} = 1\,400\,\mathrm{W{\cdot}m^{-2}}$. Further assume that the Hubble Space Telescope i also located at a distance of L from P5. Do not consider the absorption in the interplanetar medium, assume the isotropic scattering of photons by P5 and do not take into account th photons absorbed and subsequently radiated back by P5.

Lukáš read some stuff about exoplanets

Problem FoL.28 ... weird gravity

magine that you are standing on the inner side of the wall of a huge open-ended cylinder wit
base radius of $R = 1000\,\mathrm{m}$*, The cylinder is rotating about its axis with a constant angula*
peed so that the centrifugal acceleration you can feel is the same as the acceleration due t
ravity on the Earth's surface $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*. The cylinder itself is virtually massless an*
t is placed in vacuum, outside the influence of gravitating bodies. Imagine you throw a ba
traight up from the surface you are standing on, giving it an initial speed of $v = 10\,\mathrm{m{\cdot}s^{-1}}$ *i*

direction perpendicular to the surface. Compute how far from you the ball lands. The distanc
s measured along the surface, in the frame of reference connected with the rotating cylinde
State your answer in metres. *Kuba whirling around*

Problem FoL.29 ... gramme of hexagram

Consider a fractal which resulted from infinite plunging of hexagrams into each other (see th igure). Only the coloured parts have mass and their total mass is exactly one gramme. Th massive parts are made of homogeneous material and the radius of the fractal is one centimetr the radius is measured from the centre to the furthermost vertex of the fractal). What is th moment of inertia with respect to the axis perpendicular to the plane of the fractal and goin hrough the centre of the fractal? The result should be stated in terms of $\mathrm{g{\cdot}cm^2}$*. The momen f inertia of an equilateral triangle with a side lenght of* a_t *and mass* m_t *with respect to th xis perpendicular to the plane of the triangle and going through its centre of mass is*

$$I_\mathrm{t} = \frac{1}{12} m a_\mathrm{t}^2 \,.$$

Karel was thinking about moments of inertia

Problem FoL.30 ... crystal mirror

Consider an aquarium suspended in the air. We can assume that it is infinitely big, so no matte from which direction the rays are coming, they always hit a wall (see the figure). The wall are made of glass with a refractive index of $n_1 > 1$. The aquarium is filled with an unknow transparent liquid with a refractive index of $n_2 > 1$. The refractive index of air is $n = 1$. Ther is a sheet of paper placed under the aquarium with the solution to this problem written dow on it. Find the smallest refractive index of the liquid so that we would not be able to see th paper from aside of the aquarium (see the figure).

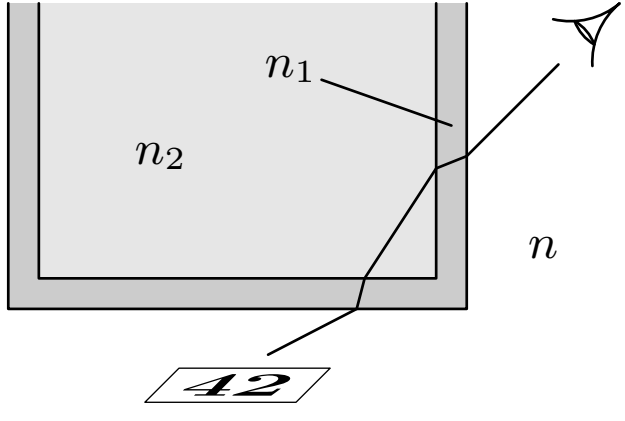


Fig. 5: To the problem 30

f(Aleš) brought up a problem during the brainstorming

Problem FoL.31 ... running mirror

Consider a system containing a converging lens with a focal lenght of $f = 20\,\mathrm{cm}$ and a movabl convex mirror with a radius of curvature of $3f$ (see the figure). At $t = 0$, when the mirro and the lens are in contact, we start to move the mirror with a speed of $v = 1\,\mathrm{m{\cdot}s^{-1}}$ awa from the lens. What should be the position of the object as a function of time, in order for it image to stay at a distance of $2f$ leftwards from the lens? Assume that you can write $x(t) =$ $= f \cdot (v^2t^2 + vft - 3f^2)/(v^2t^2 - A)$ for the position of the object. Determine the constant A Assume that the speed of light is infinite and that you can use the paraxial approximation.

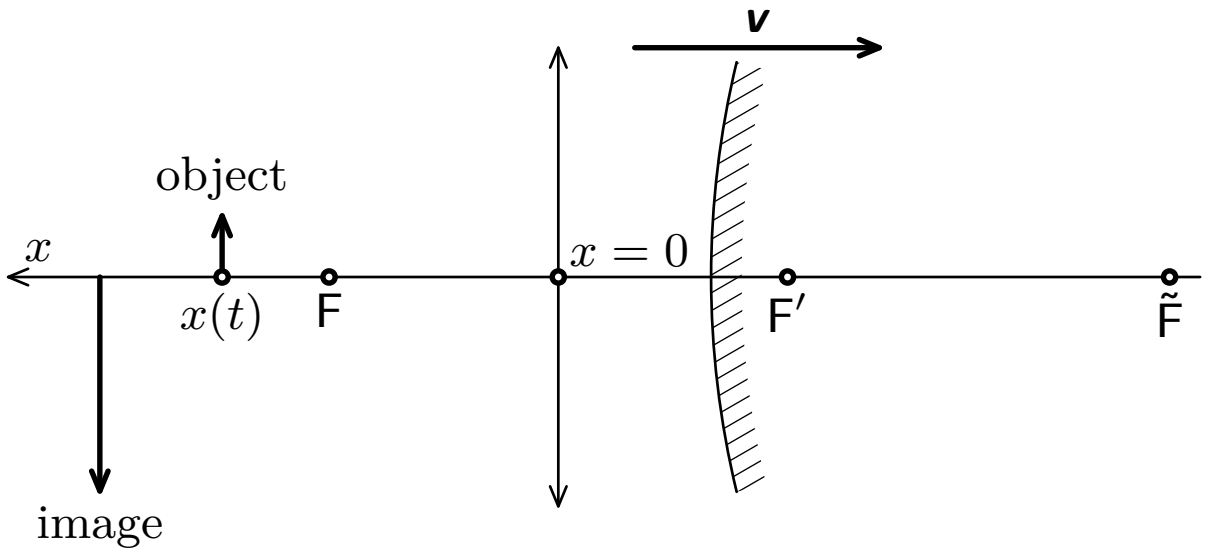


Fig. 6: To the problem 31

Lukáš, sitting on an optical bench

Problem FoL.32 ... charged robber

A robber weighing $m = 50\,\mathrm{kg}$ was running away from the policemen. He decided to save himse by jumping from a cliff. Luckily enough, he stole a charge of $q = 10\,\mathrm{C}$. There is a homogeneou magnetic field below the cliff, reaching the height $a = 10\,\mathrm{m}$. The field is perpendicular to th vertical and its magnitude is $B = 10\,\mathrm{T}$. What is the maximum height of the cliff h (in metres

rom which the robber can jump without reaching the surface? Do not take into account th ir resistance. The acceleration due to gravity is $9.81\,\mathrm{m{\cdot}s^{-2}}$ *and the velocity of the robber a he moment of entering the magnetic field has a vertical component only. Assume that th magnetic field is pointing in the right direction so the robber does not hit the cliff.*

From the head of Tomáš E

Problem M.1 ... drive safely!

Two cars with different types of tyres are decelerating in summer on a straight dry road from a velocity $v_0 = 100\,\mathrm{km{\cdot}h^{-1}}$. *Initially, they are riding next to each other and start braking at the same time. When the car with summer tyres comes to the rest the other one, with winter tyres, is still moving with a velocity* $v_1 = 37\,\mathrm{km{\cdot}h^{-1}}$ *and stops after 6 more meters. What is the fraction of the horizontal deceleration of the car with winter tyres to the horizontal deceleration of the other car?*

Michal heard this on radio

Problem M.2 ... wheelspokes

Consider an eight-spoked wheel with radius $r = 30\,\mathrm{cm}$ *rotating with angular speed* $\omega = 2.5\pi\,\mathrm{rad\cdot s^{-1}}$ *about a fixed axle through its centre. There are some boys shooting a bow in the direction of the wheel, trying to make the arrows pass freely through the gaps between the spokes. The length of one arrow is* $l = 23\,\mathrm{cm}$. *Assuming that the spokes and the arrows are negligibly thin, find the minimal speed of the arrows so that the boys would succeed in their objective. State your answer in metres per second.* *Zdeněk and his head spinning all around*

Problem M.3 ... spleen on a bridge

Bored person is standing on a bridge with the height $h = 15\,\mathrm{m}$ and is dropping pebbles on car passing along a straight road beneath him. In a distance he spots an approaching motorcyclis and decides to hit him. He estimates the motorcyclists's instantaneous velocity to be v = $= 72\,\mathrm{km{\cdot}h^{-1}}$ and his horizontal distance $d = 500\,\mathrm{m}$. He calculates when he should drop th pebble and he indeed drops it at the calculated moment. However, when the pebble hits th ground the motorcyclist is already $x = 50\,\mathrm{m}$ behind the intended point of collision. What i the difference, in kilometers per hour, between the estimate of the motorcyclist's velocity an his actual velocity if we assume that the initial horizontal distance is guessed precisely and th velocity was constant throughout the motion? *Kiki, during a stroll in Brno*

Problem M.4 ... broken altimeter

A curious sky observer notices an airplane which is approaching in a way that it eventuall passes exactly above his head. At one point, when the airplane is still approaching, the observe measures that it is $\alpha_1 = 1.3\,\mathrm{rad}$ *above the horizon. However, a noise from its engines is comin from a direction* $\alpha_2 = 0.5\,\mathrm{rad}$ *above the horizon. The observer measures an angular velocity o the airplane at the moment when it is passing above his head* $\omega = 0.125\,\mathrm{rad{\cdot}s^{-1}}$*. Based on thes inputs calculate the height* h *of the airplane, assuming it is constant throughout the motion The speed of sound is* $c = 340\,\mathrm{m{\cdot}s^{-1}}$ *and unvarying with the altitude. Neglect the finite spee of light.*

Kuba was crossing half of the globe

Problem E.1 ... a different one

Filip, who is colour blind and equipped with a red He–Ne *laser (providing light with wavelen*
gth $\lambda_1 = 633\,\mathrm{nm}$*), decided to measure the refractive index of his little glass made of borosilicat*
glass (designated as BSC7) for the wavelength of his laser (corresponding to red colour). Th
method used was to measure the critical angle of refraction of the laser beam incident on glass
vacuum interface and to infer the refractive index thereof. However, because of well-sustaine
ness in the container where he stores his lasers, he used a green laser (wavelength $\lambda_2 = 555\,\mathrm{nm}$
instead of the red one, by mistake. Find the fractional error of his result in terms of permill
(parts per thousand) assuming that the measurement was not subject to other errors of an
kind. For the BSC7 glass we have the respective refractive indices for the wavelengths $\lambda_1 =$
$= 633\,\mathrm{nm}$ *and* $\lambda_2 = 555\,\mathrm{nm}$ *equal to* $n_1 = 1.51508$ *and* $n_2 = 1.51827$*.*

Honza stumbled upon while in optics lab

Problem E.2 ... firefly

A neon lamp is connected through a resistor to a rigid source of alternating voltage of a roo
mean squre voltage 230 V *and frequency* 50 Hz. *Its ignition voltage (the striking voltage) is* 120 V
and the maintaining voltage is 80 V. *How long it will stay lit during one half-period? Assum*
that all resistors in the circuit are such that you do not have to take drop in the current int
consideration. Please provide the result in ms.

f(Aleš) wanted to read in the evening but he didn't have any lamp

Problem E.3 ... shut it down!

A square elevator with mass $m = 1000\,\mathrm{kg}$ *and a side length of* $a = 3\,\mathrm{m}$ *is moving in a very lon square shaft of a side length* $b = 4\,\mathrm{m}$ *with a speed of* $v = 2\,\mathrm{m{\cdot}s^{-1}}$*. Since this is the experimenta physics building the elevator has a point charge* $q = 1\,\mathrm{C}$ *embedded in the middle of its floor. Th most problematic lab just created a strong homogenous electric field with potential differenc between the walls of the shaft* $U = 1000\,\mathrm{V}$*. The electric field is perpendicular to the motion o the elevator and also perpendicular to the walls of the shaft. The suspension of the elevator i so long that you can assumme that it moves along a straight line. What is the maximal tim for which the electric field can last so the elevator still does not hit the wall of the shaft durin the field's action?*

(Aleš) had an afternoon filled with thoughts about elevators and electricity kept meddling into i

Problem E.4 ... anullator

Consider two long parallel conductors $\{a\}^{\{a\}}$ *metres apart, where* $\{a\}$ *denotes the numerica value of a physical quantity* a*, so* $a = \{a\}[a]$*, where we write* $[a]$ *for the unit of* a*. Both conductor are placed in vacuum and the currents through them flow in the directions opposite to on another. Assume that a current of* $I_1 = 1\,\mathrm{A}$ *flows through the conductor nr. 1. Find the curren*

lowing through the conductor nr. 2 given that the magnetic field is zero at a perpendicula
listance of $\{b\}^{\{b\}}$ *from the conductor nr. 1 (that one which is further from the conductor nr*
?). Further assume that the distance a *is* $30\,277\,604\,100\,\mathrm{m}$ *longer than one astronomical unit*
hat the light travels through the distance b *in 10 minutes and that the astronomical unit i*
precisely $149\,597\,870\,700\,\mathrm{m}$. *f(Aleš) wrote a sweet dou*

Problem X.1 ... in the subway

There are two escalators, one is leading out of the subway station, the other one into it. The
move in the opposite dirrections with velocities $0.5\,c$*, where* c *is speed of light in vacuum. Bo*
s in a hurry to get to the platform and he's running with velocity $0.6\,c$ *with respect to th*
scalator. Bobek, on the other hand, is trying to get out of the subway with velocity $0.4\,c$
elated to escalator. We are the observers standing on the platform. What is the velocity o
heir mutual movement according to our measurement?

Dominika took a subway for the first time in her life

Problem X.2 ... crash

A proton with invariant mass of $938\,272.0\,\mathrm{keV}$ *and kinetic enegry of* $1\,\mathrm{MeV}$ *hits a nucleus of th*
sotope $^{7}_{3}\mathrm{Li}$ *with mass of* $7.016003\,m_u$ *and induces a decay to two non-excited* α *particles wit*
nvariant mass of $3.727379\,\mathrm{GeV}\cdot c^{-2}$*. What will be the total kinetic energy in* MeV *of these tw*
particles? Consider $m_u = 931.2720\,\mathrm{MeV}\cdot c^{-2}$ *and* $c = 299\,792\,458\,\mathrm{m}\cdot\mathrm{s}^{-1}$*.*

Even f(Aleš) used to play marbles

Problem X.3 ... blip

A particle has a mass increased sevenfold from its invariant mass due to its movement. Yo
an follow the particle on a path $l = 1\,\mathrm{m}$ *long with a measuring device. How fast must th*
levice be to register the particle, i. e., what is the shortest time interval it has to distinguis
n order to register the particle? The speed of light is $c = 299\,792\,458\,\mathrm{m{\cdot}s^{-1}}$*. State the resul*
n ns.

f(Aleš) writing when healthy, writing when sick

Problem X.4 ... cracking right now

An unknown sample of rock contains 2 % *of thorium, which contains* 0.05 % *of the* $^{232}_{90}$Th *radionuclide. The weight of the sample is* 100 g*. What is the activity of this sample if the half-life of thorium is* $1.4 \cdot 10^{10}$ years*? Assume that nothing else decays except the mentioned thorium.*

f(Aleš) recalled the loading of equipment for a camp